Inheritance patterns & the pedigree
A published Indian case: the pedigree pointed to recessive inheritance

Summary for doctors and healthcare teams, written in our own words from a published, peer-reviewed case report. Please read the original article (link below).

Source: Patil PP, Rajagara S, Dhamankar S, Yalamgonde M, Thakare P. *Congenital leptin receptor deficiency: A novel leptin receptor gene mutation in an Indian family producing severe early-onset monogenic obesity.* Journal of Pediatric Endocrinology and Diabetes. 2023;3(1):29–33. doi:10.25259/JPED_5_2023. SRCC–Narayana Health Hospital and Seth V.C. Gandhi & M.A. Vora Municipal General Hospital, Mumbai. Open access (CC BY-NC-SA 4.0). ispae-jped.com

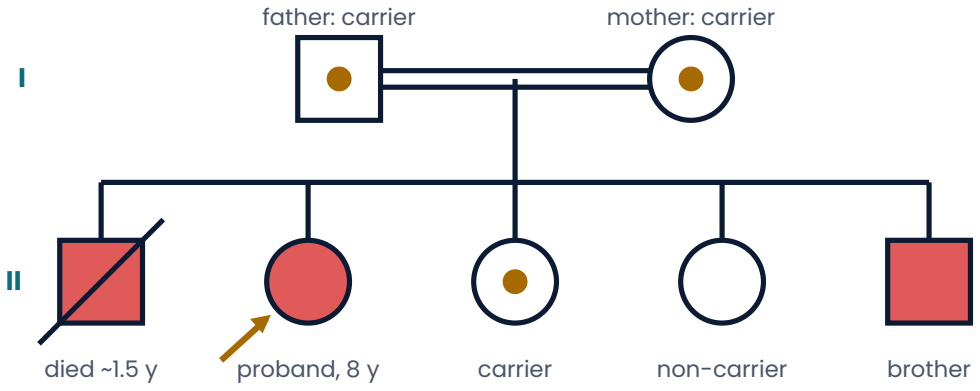
The patient

- Girl, 8 years; second of five children
- Normal birth weight (~3 kg); excessive weight gain from 3–6 months
- 10.3 kg at 3 months, 19 kg at 11 months; hyperphagia
- Two bariatric surgeries (11 months and 7 years) without a favourable outcome
- Normal development; no dysmorphism

Family

- Consanguineous, healthy parents
- Younger brother: severe early obesity (18 kg at 1.5 years), hyperphagia, repeated respiratory admissions
- Elder brother: severe obesity; died at ~1.5 years of a respiratory infection
- Two unaffected sisters

Pedigree (simplified; birth order of the unaffected sisters is illustrative)



Result

Serum leptin was markedly raised. Whole exome sequencing (ACMG interpretation) found a novel **homozygous LEPR variant c.1752G>A (p.Lys584=)** (NM_002303.6) in the affected children, classified as pathogenic by the authors. Sanger testing showed **both parents heterozygous carriers**; one unaffected sister was a carrier and the other did not carry the variant. Diagnosis: **congenital leptin receptor deficiency (autosomal recessive)**. The younger brother had sleeve gastrectomy at ~2 years 2 months as setmelanotide was not available in India at the time.

EPISODE 4 · QUICK GUIDE

Inheritance patterns at a glance

Pattern	Pedigree clues	Risk to offspring
Autosomal dominant	Affected in every generation; males and females equally; father-to-son transmission possible. Watch for reduced penetrance, variable expressivity, de novo cases.	Each child of an affected parent: 50%.
Autosomal recessive	Usually affected siblings with healthy carrier parents; consanguinity increases the chance.	Both parents carriers: each pregnancy 25% affected, 50% carrier, 25% neither.
X-linked recessive	Mostly males affected, through carrier mothers; never father to son (e.g. Duchenne muscular dystrophy, haemophilia).	Carrier mother: each son 50% affected, each daughter 50% carrier.
X-linked dominant	Affected father: all daughters affected, no sons.	Affected mother: each child 50%.
Mitochondrial	Passed only through the mother, never the father; severity varies.	Depends on the mother's mutation load; specialist counselling.

Three-generation pedigree: ask

- Age at onset; age and cause of death
- Miscarriages, stillbirths, infant deaths
- Consanguinity; similar illness in relatives

Standard symbols

- Square male, circle female, diamond sex unknown
- Filled = affected; central dot = carrier
- Diagonal line = deceased; arrow = proband
- Double line between partners = consanguinity

Talking points for the doctor visit

- A pedigree is free and quick, and it guides the choice of test and the counselling.
- Pedigree segregation helps interpret a VUS; carriers can be offered cascade testing.
- Send the pedigree and clinical details with the sample.

References

- Patil PP et al. J Pediatr Endocrinol Diabetes 2023;3(1):29–33. doi:10.25259/JPED_5_2023
- Bennett RL et al. Standardized human pedigree nomenclature. J Genet Couns 2008; update on sex and gender inclusivity, 2022 (National Society of Genetic Counselors).

Watch the episode: healthtestsimplified.netlify.app/#g004 · Facebook & YouTube: Health Test Simplified